

GreyNOC EDC Operator Guide

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Plain-English Summary

This guide helps operators turn an Android phone into a safe, no-root GreyNOC everyday-carry technical workspace. Android stays installed. The phone is not rooted, unlocked, flashed, or converted into a custom operating system. Instead, the operator installs Termux from F-Droid, creates a Debian Linux environment using PRoot Distro, and adds a small set of defensive tools for administration, notes, SSH, network troubleshooting, scripting, and authorized lab work.

Best-Practice Recommendation

Do not flash random phone images, unlock bootloaders, or replace Android unless the device vendor and operating system project explicitly support that exact model. For most operators, the safer starting point is a no-root Termux and Debian workspace inside Android.

1. Scope and Responsible-Use Notice

This guide is written for public education, field readiness, and legitimate technical operations. It is intended for operators, students, technicians, SOC learners, home-lab builders, and small-business defenders who want a portable command-line environment on Android.

Use all tools only on devices, networks, accounts, and lab environments that you own or have explicit permission to administer or test. Unauthorized scanning, credential attacks, exploitation, interception, bypassing access controls, or testing third-party systems without written permission may violate law, policy, and service terms.

This guide does not provide exploit instructions, credential theft guidance, stealth techniques, malware, unauthorized access workflows, or phone-flashing instructions.

2. What This Guide Installs

Item	Recommendation
Installed approach	Android + Termux + PRoot Distro + Debian
Root required	No
Bootloader unlock required	No

Item	Recommendation
Android remains installed	Yes
Primary purpose	Portable Linux shell, SSH, scripting, documentation, and defensive troubleshooting
Cybersecurity use	Authorized labs, home networks, CTF-style practice, and defensive learning
Not included	Custom ROM flashing, packet injection, monitor mode setup, exploit chains, persistence, or bypass procedures

Termux provides the Android terminal environment. PRoot Distro provides a way to install and manage a Linux userland inside Termux. Debian provides a stable base for Linux tools and scripting.

3. Operator Requirements

Requirement	Recommendation
Android device	A phone or tablet you control and are allowed to modify at the app level
Storage	8 GB free minimum; 20 GB or more recommended
Network	Stable Wi-Fi for package downloads
Power	Keep the device charged or plugged in during installation
Keyboard	Optional Bluetooth keyboard for easier field use
Account safety	Use trusted app sources and avoid random APK download sites
Data safety	Back up important phone data before experimenting

4. Install the Base Tools

Step 1: Install F-Droid

Install F-Droid from the official F-Droid website. F-Droid is used because the actively maintained Termux package is distributed there.

Step 2: Install Termux from F-Droid

Open F-Droid, search for Termux, and install it. Avoid outdated Termux builds from random APK sites.

Step 3: Grant storage access

Open Termux and run:

```
termux-setup-storage
```

When Android asks for permission, tap Allow.

Step 4: Update Termux

```
pkg update && pkg upgrade -y
```

Step 5: Install required Termux packages

```
pkg install proot-distro git wget curl nano openssh -y
```

5. Install Debian Inside Termux

Step 6: Install Debian

```
proot-distro install debian
```

Step 7: Log into Debian

```
proot-distro login debian
```

Step 8: Update Debian

Run this from inside the Debian shell:

```
apt update && apt upgrade -y
```

Step 9: Install everyday Linux tools

```
apt install sudo git curl wget nano vim python3 python3-pip dnsutils whois  
netcat-openbsd openssh-client -y
```

Step 10: Create a normal operator user

Replace `greyuser` with your preferred username.

```
adduser greyuser
usermod -aG sudo greyuser
```

Exit Debian:

```
exit
```

Log back in as the new user:

```
proot-distro login debian --user greyuser
```

6. Create the GreyNOC EDC Workspace

Inside Debian as the normal user, create a clean folder structure:

```
mkdir -p ~/greynoc-edc/{notes,scripts,logs,reports,clients,lab,tools}
mkdir -p ~/greynoc-edc/reports/{templates,evidence,completed}
touch ~/greynoc-edc/notes/field-notes.md
touch ~/greynoc-edc/logs/session-log.txt
```

Recommended folder use:

Folder	Purpose
<code>notes</code>	Operator notes, checklists, reminders
<code>scripts</code>	Small helper scripts
<code>logs</code>	Local session notes and command output from authorized work
<code>reports</code>	Report templates, evidence copies, completed reports
<code>clients</code>	Public-safe client folders; do not store secrets in plain text
<code>lab</code>	Home-lab and training artifacts
<code>tools</code>	Operator utilities and helper files

7. Create a Quick Launch Command

From regular Termux, create a small helper command so the operator can start Debian by typing `linux-phone`.

First exit Debian if you are inside it:

```
exit
```

Then run this in Termux. Replace `greyuser` with the username you created:

```
echo 'proot-distro login debian --user greyuser' > $PREFIX/bin/linux-phone  
chmod +x $PREFIX/bin/linux-phone
```

After this, start the Linux workspace with:

```
linux-phone
```

8. Operator Baseline Toolset

Inside Debian as the normal user, install a practical defensive baseline:

```
sudo apt update  
sudo apt install -y  
git curl wget nano vim tree unzip zip  
python3 python3-pip python3-venv  
dnsutils whois netcat-openbsd  
openssh-client jq htop tmux
```

This baseline supports SSH, scripting, DNS lookups, basic troubleshooting, file handling, and repeatable terminal sessions.

9. Create a Status Script

Create a simple operator status script:

```
nano ~/greynoc-edc/scripts/status.sh
```

Paste:

```
#!/usr/bin/env bash  
  
echo "=== GreyNOC EDC Status ==="  
date  
echo  
echo "User: $(whoami)"  
echo "Host: $(hostname)"
```

```
echo
echo "Python:"
python3 --version
echo
echo "SSH:"
ssh -V
echo
echo "Network:"
ip addr 2>/dev/null | head -40
```

Save, then make it executable:

```
chmod +x ~/greynoc-edc/scripts/status.sh
```

Run it:

```
~/greynoc-edc/scripts/status.sh
```

10. Optional Authorized Lab Tools

This section is only for authorized labs, home networks, classroom environments, capture-the-flag practice, and defensive learning. Install only what you understand and only use these tools where you have permission.

```
sudo apt update
sudo apt install nmap tcpdump tshark nikto gobuster sqlmap -y
```

Important Limitation

Some tools may install but not work like they do on a laptop. Android, PRoot, and phone hardware can limit raw sockets, monitor mode, packet injection, kernel modules, background services, USB device access, and privileged network operations.

For wireless research, most phone Wi-Fi chipsets do not support monitor mode or packet injection in a standard Android setup. External adapters, root access, kernel support, and legal authorization are usually required. Those steps are outside the scope of this public no-root guide.

11. Daily Operator Use

Action	Command or Step
Start Linux	Open Termux and run <code>linux-phone</code>

Action	Command or Step
Exit Linux	Run <code>exit</code>
Update Debian	Run <code>sudo apt update && sudo apt upgrade -y</code> inside Debian
Install a package	Run <code>sudo apt install package-name -y</code> inside Debian
Check Python	Run <code>python3 --version</code>
Check SSH	Run <code>ssh -V</code>
Check IP addresses	Run <code>ip a</code> inside Debian or <code>ip addr</code> in Termux
Run status check	Run <code>~/greynoc-edc/scripts/status.sh</code>

Quick reference session:

```
linux-phone
sudo apt update && sudo apt upgrade -y
python3 --version
ssh -V
~/greynoc-edc/scripts/status.sh
exit
```

12. Field Notes and Evidence Hygiene

Operators should write down what they do, when they do it, and why. Keep notes professional, factual, and permission-aware.

Recommended note format:

```
Date:
Operator:
Authorized environment:
Purpose:
Commands or actions performed:
Observed result:
Follow-up needed:
Sensitive data captured? yes/no
Sanitized copy created? yes/no
```

Do not store credentials, private keys, client secrets, API tokens, production logs, or sensitive screenshots in plain text on the phone. If a file is intended for public sharing, make a sanitized copy, remove metadata where appropriate, and review visible clues before publishing.

13. What This Setup Is Good For

This setup is good for:

- SSH into authorized systems
- Linux command practice
- Python scripting
- DNS and whois lookups
- Documentation and field notes
- Small report drafts
- Home-lab troubleshooting
- SOC learning and defensive workflows
- Carrying offline checklists and runbooks

14. What This Setup Is Not

This setup is not:

- A replacement for a dedicated Linux laptop
- A full custom Android ROM
- A rooted phone environment
- A packet-injection platform
- A stealth or bypass toolkit
- A guarantee that every Linux package will work
- A safe place to store unencrypted client secrets

Full desktop environments may be slow, battery-heavy, and hard to use on a small screen. Systemd-based services may not behave the same way inside PRoot. Low-level network testing can be blocked by Android permissions, kernel limits, or hardware limits.

15. Troubleshooting

Problem	Recommended Action
<code>pkg update</code> fails	Check Wi-Fi, rerun <code>pkg update</code> , then try <code>pkg upgrade -y</code> again
<code>proot-distro install debian</code> fails	Check storage and network stability; remove broken install with <code>proot-distro remove debian</code> , then reinstall
<code>sudo</code> does not work	Make sure you created a normal user and added it to the sudo group
Permission denied	Confirm whether the command belongs in Termux or Debian; use <code>sudo</code> only inside Debian after user setup
Command not found	Install the package or confirm you are in the right environment
Phone gets hot	Pause package installs, plug in the device, close other apps, and avoid long compile jobs

Problem	Recommended Action
Storage fills up	Remove unused packages and old lab artifacts; move non-sensitive archives off-device
Tool works on laptop but not phone	Assume Android, PRoot, kernel, or hardware limits may be blocking the feature

Helpful commands:

```
proot-distro list
proot-distro login debian
proot-distro remove debian
sudo apt update
sudo apt autoremove -y
```

16. Operator Safety Rules

1. Get permission before touching a network or system.
2. Keep client, employer, and production data off public posts.
3. Use synthetic examples for screenshots and demonstrations.
4. Do not store secrets in plain text.
5. Keep the phone updated.
6. Use screen lock and device encryption.
7. Separate personal, lab, and client materials.
8. Record what you did in clean field notes.
9. Delay or avoid posts from live work, travel, incidents, or sensitive locations.
10. When unsure, stop and ask for authorization.

17. Public Posting and Sharing Rules

Before posting screenshots, videos, guides, or operator notes, check for:

- IP addresses
- hostnames
- usernames
- emails
- tickets
- customer names
- credentials
- API keys
- browser tabs
- terminal paths
- Wi-Fi names
- location clues
- badge IDs

- license plates
- faces
- reflections
- document metadata

Always work on a copy. Sanitize the copy. Verify the copy. Publish only the public-safe copy.

18. Release Checklist for This Guide

Before publishing this guide externally:

Check	Status
Internal-only references removed	Pending review
No exploit instructions included	Pending review
No customer environments included	Pending review
No credentials, keys, or live targets included	Pending review
Commands tested on a clean Android/Termux setup	Pending review
Safety and legal language reviewed	Pending review
Formatting, links, and accessibility checked	Pending review
Final PDF or web version archived	Pending review

19. References for Operators

Use official project documentation when validating installation steps and package behavior:

- Termux project site
- Termux on F-Droid
- Termux PRoot documentation
- PRoot Distro project
- Debian package documentation
- Android device security settings
- GreyNOC public safety and operator guidance

Legal and Safety Disclaimer

This document is for education, defensive operations, and authorized administration only. GreyNOC does not endorse unauthorized access, scanning, exploitation, credential attacks, interception, evasion, or bypassing access controls. Operators are responsible for following applicable laws, organizational policy, platform terms, and written authorization requirements.